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Perivascular spaces in the brain: anatomy, physiology and pathology. [Au: Limit for the title is 90 characters, including spaces, so I have shortened it. OK? YES Please feel free to suggest an alternative within the limit.]

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Abstract

Perivascular spaces include a variety of passageways around arterioles, capillaries and venules in the brain, along which a range of substances can move. Although perivascular spaces were first identified over 150 years ago, they have come to prominence recently owing to advances in knowledge of their roles in clearance of interstitial fluid and waste from the brain, particularly during sleep, and in the pathogenesis of small vessel disease, Alzheimer disease and other neurodegenerative and inflammatory disorders. Experimental advances have facilitated in vivo studies of perivascular space function in intact rodent models during wakefulness and sleep, and MRI in humans has enabled perivascular space morphology to be related to cognitive function, vascular risk factors, vascular and neurodegenerative brain lesions, sleep patterns and cerebral haemodynamics. Many questions about perivascular spaces remain, but what is now clear is that normal perivascular space function is important for maintaining brain health. [Au: I have removed a few sentences from here that I thought didn't really tell the reader much. I think instead it would be of more value to expand on the following sentence to provide more specifics about what is covered in the Review, particularly in relation to MRI, as this makes up a large part of the article but is not really mentioned. Please add.] Here, we review PVS anatomy, physiology and pathology, particularly as seen on magnetic resonance imaging in humans, while translating from-between models to-and humans to, highlighting knowns, unknowns, controversies, and clinical relevance.

[H1] Introduction

The spaces that surround small blood vessels in the brain are variously known as perivascular or paravascular spaces, [Au: Given that the abbreviation is sometimes used for the plural and sometimes for the singular and that we generally avoid abbreviating 2-word terms, I suggest the abbreviation is not used. OK? YES] including periaarteriolar, pericapillary and perivenular spaces. Although these spaces were described in the human brain over a century ago, they have come to prominence in the past decade owing to advances in the sensitivity of in vivo visualization tools, such as MRI in humans and 2-photon imaging [Au: We prefer not to abbreviate 2-word terms unless the abbreviations are very familiar, so I suggest leaving 2-photon imaging in full throughout OK] via cranial windows in rodents. These modalities provide opportunities to understand the physiology, complex nature and importance of the brain's fluid and waste clearance systems and how — as part of these systems — [Au: Addition made to clarify how perivascular spaces relate to the other drainage systems discussed. OK? YES] perivascular spaces influence the pathogenesis of common cerebrovascular, neuroinflammatory and neurodegenerative disorders.

[Au: The introduction cannot be broken into subsections, so I have removed the subheadings. OK] Converging information from human studies and rodent models suggests that perivascular spaces contribute to the maintenance of brain health and that dysfunction of perivascular spaces contributes to common neurological disorders. [Au: Although informative, I think the passage that provided an overview of the associations and findings for perivascular spaces was too detailed for the introduction, which is intended to provide a brief background and context for the main, detailed parts of the article. OK]

However, many aspects of perivascular spaces have been, and remain, controversial since these spaces were first described in the mid-1800s²⁰ (Box 1)^{21,22}. Current debates relate to whether their reported associations with vascular risk factors, neurological diseases and small vessel disease (SVD)^{1,5} are accurate and important, [Au: OK? YES] or whether perivascular spaces are an epiphenomenon. [Au: Sentence edited to be more specific about what the debates relate to, and emphasize that their being an epiphenomenon is an alternative. Does the sentence retain your meaning? YES] Previously, perivascular spaces ~~were thought to have been attributed to~~ histopathological fixation artefacts or ~~a thought to result of—from~~ brain tissue loss during ageing, so have been overlooked. Other controversies are the anatomical structure of perivascular spaces²³, their relationships to arterioles, capillaries and venules²⁴, their connections with

cerebrospinal fluid (CSF) compartments²⁵, their relationships to meningeal lymphatic drainage channels²⁶⁻²⁸, the direction of fluid drainage from the brain, the role of aquaporin 4 (AQP4) in this process²¹, and the function of perivascular spaces during sleep^{12,29}. [Au: The statement about the possible methodological reasons for these controversies is not necessary in the introduction.OK]

In this Review, we discuss what is known and not known about perivascular spaces by considering their anatomy, physiology and roles in pathology. We first discuss clinical evidence for the importance of perivascular spaces based on MRI and supporting data from histopathology, and subsequently consider information obtained from preclinical studies about perivascular space structure, their relationships with other fluid drainage systems [Au: Addition made to make clear that you will discuss the other systems as well. OK?YES] and their functions in health and disease models. Throughout, we aim to highlight major controversies and gaps in knowledge (Box 1) to indicate where further research is needed.

For the purposes of this Review, we use the term 'perivascular space' broadly to refer to any potential passageway that allows fluid to move along the outside of a vessel, [Au: This wording in your rebuttal made very clear how broadly you are applying the term, so I felt inclusion of this wording was helpful in the manuscript. OK?YES] including the small spaces that are visible in the brain on MRI or at post-mortem that run into the brain in the same direction as perforating vessels and that are thought to be contiguous with the pericapillary potential spaces³⁰. Given that the dynamics of perivascular spaces are so important, we focus on in vivo data and the latest methods that enable detailed in vivo analysis of perivascular spaces^{31,32}. [Au: The sentence about the potential for biomarkers did not seem appropriate here, so I have suggested it is removed. OK?YES]

[H1] A brief history

Perivascular spaces were originally described in 1843 by Durand Fardel and in 1849 by Pestalozzi, but are often ascribed to Rudolf Virchow and Charles Robin, who described spaces around brain perforating vessels in 1851 and 1859, respectively²⁰. Perivascular spaces have since been referred to as Virchow–Robin spaces, ~~but despite~~ these two experts ~~disagreed-disagreeing~~ on whether perivascular spaces connect with the subarachnoid space and whether perivascular spaces are a type of 'brain lymphatic'. Robin proposed that perivascular spaces connect with perineuronal spaces²⁰ ~~but this~~ property which is now recognized as ~~one-part~~ of lymphatic ~~vessels~~^{8,17,25} drainage channels^{8,17,25}. [Au: Edited for clarity. Meaning correct?ALMOST – NOW OK] In 1843, [Au: 1843 is earlier than the date given

above for the original description of perivascular spaces – please clarify.**YES**
GOOD POINT; I THINK DURAND FARDEL SHOULD BE MENTIONED EARLIER
SO HAVE DONE SO Durand Fardel had described enlargement of periarteriolar spaces in post-mortem brains ~~was described by Durand Fardel and, who~~ referred to their appearance in basal ganglia as 'état crible'³⁵. ~~This appearance was accompanied by abnormal arteriolar walls and perivascular inflammatory cell infiltration. By~~ In the 1950s, ~~perivascular space enlargement was noted as being pathological, accompanied by perivascular inflammatory cell infiltration and these arteriolar and periarteriolar space appearances were clearly linked with pathological arteriolar morphologies consistent with arteriolosclerosis and fibrinoid necrosis described by Miller Fisher~~³⁶. **[Au: Changes to sentence OK? NEARLY – HOPE OK AS EDITED; ALSO THOUGHT WORTH MENTIONING FISHER AS ANOTHER GIANT OF PATHOLOGY AND SMALL VESSEL DISEASE]**

Some of the original theories about the function of perivascular spaces derived from rodent experiments that were conducted in the early 1900s to investigate CSF production and circulation. These experiments showed that Prussian blue injected into the subarachnoid space entered the perivascular spaces^{37,38} ~~suggesting a potential interstitial fluid exchange role for CSF.~~ **[Au: Please add a sentence to explain what these experiments were taken to mean about perivascular space function.OK]** Furthermore, multiple experiments in the 1920s³⁹ suggested that perivascular spaces extend along arterioles, capillaries and venules, and communicate freely with perineuronal spaces and other spaces between glial elements and fibre tracks. These conclusions were based on studies in which dyes were injected into CSF alongside intravenous injection of hypertonic saline to increase dye uptake, but the saline also caused tissue shrinkage and this confounding factor might have contributed to the idea that perivascular spaces were merely fixation artefacts. **[Au: The comment about the use of hypertonic saline was interesting but seems to be more of an aside, so I think for brevity it is better to remove it. OK?WELL, I THINK IT IS BETTER AS IT IS; THERE IS A PATENT ON USE OF HYPERTONIC SOLUTIONS TO INCREASE PVS FLUID UPTAKE AS A POTENTIAL TREATMENT FOR AD, SO BETTER NOT OMITTED ALTOGETHER]** Subsequent experiments were designed to address this controversy by isolating the effects of the hypertonic saline²⁰ and confirmed that dyes injected into the CSF reached perivascular spaces, particularly those in the basal ganglia. **[Au: I have substantially cut down the description of the India Ink/colloidal carbon experiments, because I think the description was not clear and additional explanation would have made this passage longer than justifiable. I therefore**

think an overall summary of the main points is more useful to the reader here.

OK?OK A space observed outside the pial membrane was considered to be an artefact because it only occurred with use of hypertonic saline²⁰; this finding might be a source of the persisting idea that histologically observed and MRI-visible perivascular spaces are artefacts.

The advent of widespread MRI use in the 1980s, particularly T2-weighted sequences, enabled detection of perivascular spaces as **[Au: OK?OK AS AMENDED]** small, linear, fluid-filled structures parallel to the known direction of perforating vessels in the midbrain, hippocampus, basal ganglia and cerebral hemispheric white matter of the centrum semiovale⁴². **[Au: I felt the information about the visibility with different MRI protocols was unnecessary here and its removal enabled the narrative to flow more smoothly. ACTUALLY MINIMAL INFO ON SEQUENCES IS IMPORTANT SINCE PVS ARE MAINLY VISIBLE ON T2, BARELY ON T1, NOT ON OTHER SEQUENCES. LACK OF REALISATION OF DIFFERENT SEQUENCES SENSITIVITIES HAS RESULTS IN UK BIOBANK SCANNING 100,000 PEOPLE BUT NOT BEING ABLE TO ASSESS PVS!]** These structures were largely ignored until the early 2000s, when several groups noted that perivascular space visibility varied widely and went on to study their clinical phenotypes and associations with vascular risk factors^{9,23,44-46}. **[Au: OK? with risk factors for what? Mainly vascular risk factors]**

Subsequently, major advances in in vivo experimental methods have accelerated research into the structure and function of perivascular spaces. These advances include the development of 2-photon imaging via cranial windows⁴⁷ in alert animals¹², dynamic MRI to track CSF fluid movements in rodents⁴⁸, **[Au: OK, to simplify? NEED TO BE CLEAR THAT IT IS CSF NOT JUST BLOOD]** mathematical modelling of fluid movement^{33,49}, **[Au: OK, to simplify? YES]** advances in analysis of microscopy and MRI data⁵⁰, and sophisticated histopathological and electron microscopy techniques. Some of these advances are now being translated into methods for in vivo human MRI^{32,51,52} with sophisticated image analysis^{13,53}. As a result, reliable information from both laboratory and human studies is now converging. Nevertheless, controversies about anatomy and physiology remain, hindering translation of the knowledge gained into clinical applications. **[Au: Addition OK, to highlight where the gaps in knowledge remain? OK]**

[H1] MRI of human perivascular spaces [Au: I have suggested an overarching heading to include the entire section on MRI. OK?OK]

[H2] Anatomy

The historical and contemporary histology-based literature on the structure of perivascular spaces, and the other spaces to which perivascular spaces connect, is complicated and confusing. Knowledge of perivascular spaces that are visible with routine clinical brain MRI is much clearer and is producing a more consistent picture with the potential for clinical application. [Au: Addition made to indicate to the reader that the MRI aspects have the potential for clinical application. OK?YES]

Perivascular spaces in some form (including potential passageways [Au: Removal of “virtual” OK? I understood this to be an alternative term to “potential” and therefore unnecessary, but please reverse if these are different- ITS OK]) are thought to surround arterioles, capillaries and venules in the brain. The perivascular spaces that are visible in the brain parenchyma on MRI run perpendicular to the brain's surface and are parallel to and spatially correlated with perforating vessels. [Au: Please cite reference(s) to support this statement REF 4, 10, 23, 42 ARE OK HERE] Therefore, it is reasonable to believe that these visible perivascular spaces are related to perforating vessels. Perivascular spaces that run along the plane of the image are seen as linear, and those perpendicular to the plane of the image are seen as dot-like (Figure 1)⁴³.

[H3] Locations in the brain

One or two small perivascular spaces are often visible on MRI in young brains, but a greater number [Au: Correct that you mean a greater number become visible rather than those already there becoming easier to see? GREATER NUMBER BECOME VISIBLE] usually become visible with increasing age^{2,8}. Regardless of [Au: OK? YES] number, perivascular spaces are typically seen in specific brain regions (Figure 1): the basal ganglia (the lentiform nucleus internal and external capsules) immediately superior to the basal perforating substance, where they often connect with the cisternal CSF (Figure 2); the centrum semiovale, running centripetally from the external aspect of the white matter towards the lateral ventricles, including in the anterior temporal poles in monogenic SVDs, such as cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy CADASIL;⁵⁴ the hippocampus; and the midbrain, pons and sometimes in the cerebellar white matter⁶.

Generally, individuals with numerous perivascular spaces in one region have numerous perivascular spaces in all typical areas. For instance, the number of basal ganglia perivascular spaces correlates highly with that of centrum semiovale perivascular spaces⁵⁵, although spaces can be more prevalent or larger in one region

than the other, and their associations with vascular risk factors, inflammatory markers and neurological diseases [Au: Associations with what? Please clarify] can differ (see below [Au: Please name the relevant section below – section headed ‘insights from MRI’ second para, and section headed ‘risk factor associations’]). Hence visual scoring methods rate perivascular spaces in several key brain regions. [Au: More detail on quantification is given in the section below, so the sentence here on this area seemed unnecessary –ACTUALLY NO, DIFFERENCES IN RISK FACTOR AND DISEASE ASSOCIATIONS ARE A KEY REASON FOR ASSESSING PVS IN DIFFERENT REGIONS]

[Au: Again, reference to quantification removed from here. OK? –NO, there are important anatomical differences to keep in mind in understanding pvs function] Regional anatomical differences in perivascular space structure also justify separate regional quantification. High-field (7T) MRI has revealed that basal ganglia perivascular spaces communicate directly with the basal subarachnoid cisterns (Figure 2)⁴, whereas perivascular spaces in the centrum semiovale, which surround vessels that enter the brain from the convexity cortex, appear to start a few millimetres beneath the cortex⁴. [Au: If I understand correctly, the place that the PVS start is what is being contrasted. I have therefore simplified to emphasize the contrast. Is this correct, or have I misunderstood? – NO, IT LOOKS LIKE PVS START BELOW THE CORTEX ON MRI BUT OTHER INFORMATION SUGGESTS THAT THEY MUST CONNECT WITH THE SUPERFICIAL BRAIN SURFACE – PLEASE LEAVE ‘APPEAR TO’] In human post-mortem tissue studies, the appearance of perivascular spaces that started immediately beneath the cortex was similar⁵⁶ ~~with to that seen on~~ lower field strength MRI (Figure 3)⁴³, the same was true in rodent models in histology and 2-photon imaging experiments⁸, demonstrating that MRI and histopathology findings correspond. [Au: Edited sentence OK? YES Addition made at the end to emphasize the agreement between MRI and histopathology. OK? YES]

[H3] *Periarteriolar, perivenular or both?*

Whether MRI-visible perivascular spaces surround arterioles, venules or both is currently under debate⁵⁷⁻⁵⁹. Most MRI at conventional field strengths cannot easily identify perforating arterioles and venules in humans. In one small study, use of 7T MRI demonstrated that MRI-visible perivascular spaces correlate spatially with arterioles but not venules⁴ (Figure 4). If images are of good quality, lower field strengths (1.5T or 3T) can be used to see perivascular spaces (with a T2 sequence) and venules (with a blood-sensitive susceptibility-weighted sequence (T2*)) in the

centrum semiovale⁶⁰, and [Au: Part of sentence removed here because it implied to me that the perivascular spaces were associated with venules. OK? OK] use of these sequences in combination suggests that the venules are distinct from the perivascular spaces (Figure 4). [Au: Please cite appropriate reference(s) ref 42 and a PAPER IS IN SUBMISSION] Of course, these findings are based on small samples, so it would be imprudent to state that all MRI-visible perivascular spaces are around arterioles. ~~In addition~~Nonetheless, ~~the main location of perivascular spaces being around arterioles is conclusion~~ would imply that ~~in general~~, visible [Au: Addition of “visible” OK?] perivascular spaces are ~~not related primarily to a sign of~~ venular dysfunction. [Au: The implication here is that perivascular spaces are related to venular dysfunction, but this is not made clear. Is that the intended meaning? If so, please clarify. Is the paragraph that follows intended as evidence that perivascular spaces are related to venular dysfunction? If so, please make this clearer.]

[Au: The first two sentences of this paragraph do not clearly relate to perivascular spaces – please clarify how they relate. Are perivascular spaces are seen alongside these pathologies? MIGHT BE EASIER IF THIS PARA CONTINUED FROM THE PRIOR ONE] ~~The presence of abnormal deep medullary venules with collagenosed walls has been described in~~ histopathological studies, ~~collagenosis of the deep medullary venules has been described in~~ the periventricular white matter alongside arteriolosclerosis in ~~old-aged~~ [Au: Can you be more specific about the age range that “older” referred to?] patients ~~aged over 60~~ in whom pre-mortem or post-mortem MRI had confirmed leukoaraiosis⁵⁷. [Au: Edited sentence OK? OK] In a patient with CADASIL, diffuse, patchy periventricular WMHs [Au: What measure of WMHs correlated? Eg. the number, the volume etc.?] ~~co-located~~~~related~~ with collagenosis of the deep penetrating venules at autopsy⁵⁹ ~~in areas where there were also numerous visible PVS~~. [Au: Please clarify how this observation relates to PVS.] However, the extent to which venous collagenosis contributes to perivascular space visibility on MRI in ~~CADASIL or~~ sporadic SVDs, ~~whether the visible perivascular spaces collocate with abnormal venules~~, and how commonly venous collagenosis occurs in monogenic SVDs such as CADASIL, are unknown and warrant investigation.

Methods are now available to visualize venules close to active lesions in multiple sclerosis⁶², where focal perivascular space dilatation has been observed during active inflammation¹⁰. These methods make it possible to see whether the visible perivascular spaces correspond with ~~abnormal~~ venules and whether similar abnormal PVS associated with [Au: From the previous sentence, I understand it's

Kommenterede [WJ1]: We are saying that the weight of evidence indicates that MRI visible pvs are NOT perivenular in location. But that we cannot assume normality of venular function. Hope clearer as reworded.

the PVS that are abnormal rather than the venules. OK? No, its potentially both venules are ever visible in vascular WMHs or in recent small subcortical infarcts in patients with sporadic or genetic SVDs. Use of this approach in humans has revealed abnormal arterioles, perhaps thrombosed, in the centre of recent small subcortical infarcts, some of which seemed to be associated with a prominent periarteriolar space⁶³, but similar venules have not been documented. Therefore, in general, the evidence supports the location of conventional MRI-visible perivascular spaces as being periarteriolar in ageing, small vessel disease and neurodegenerative disorders, and indicate development of arteriopathies and associated venular pathology. [Au: I think it would be helpful to sum up what these findings tell us about PVS location]

[H2] Quantification of perivascular spaces

Almost all MRI studies of perivascular spaces and their associations to date have relied on visual scores for quantification because, until recently, computational image analysis methods have not been sufficiently advanced to quantify such small structures (Figure 1). Several visual scoring methods have developed over the past 18 years⁴⁴, but all involve similar approaches and quantify perivascular spaces in similar brain areas⁴³. Manual counting of perivascular spaces in a scan slice is too time consuming, especially in large studies, so most scores provide qualitative estimates [Au: Addition of “provide qualitative estimates” to introduce the idea that these measures are qualitative to the reader earlier than the final sentence of the paragraph. OK? YES] of the extent of perivascular spaces on the basis of the approximate number of perivascular spaces in an anatomically defined region^{43,46,64}. Thus, fewer than 10 perivascular spaces within the basal ganglia on a defined brain scan slice ~~might-would~~ generate a score of 1, 11–20 a score of 2, 21–40 a score of 3 and >40 a score of 4⁴³. These scores are quick and practical to use in clinical research and have good reliability and repeatability⁴³, so have been applied in many individual studies, including some that have involved several thousand individuals¹. However, such qualitative scores are relatively insensitive and are limited by floor and ceiling effects.

Advances in isotropic 3D MRI acquisition and computational image analysis methods have enabled computational quantification of perivascular spaces^{3,32,65}. These methods require further testing to confirm their value, [Au: Addition made to avoid implying that they require additional testing of the patient. OK? ok] but ~~promise-are likely~~ to improve sensitivity to changes. Some methods enable several characteristics of perivascular spaces to be quantified in addition to number [Au:

Change to “number” OK? Frequency could be misinterpreted as regularity in space ok (for example, total volume of PVS, individual –sizes, lengths, widths, sphericity, directionality and proximity to other structures) and generate spatially correlated measures of tissue integrity³². Early studies in humans **[Au: Studies in humans? Please clarify]** with these methods have shown high agreement between visual scores of perivascular spaces and computational counts, volumes and individual sizes of perivascular spaces (Figure 1)³².

As yet, no equivalent method for quantifying perivascular spaces in human tissue sections is available, although similar approaches could be applied in this context and in rodents. However, dynamic MRI methods for tracking uptake and distribution of CSF tracers in perivascular spaces in rodents are much more advanced than what is currently possible in humans^{33,48}. These methods employ optimal mass transport (OMT), a mathematical method that quantifies the movement of a substance or objects through a volume **[Au: Definition removed because this term is explained in the glossary. OK? Ok, but I thought a little explanation would help]** . **[Au: The technical details of the method were not very clear and I think are unnecessary to make the main point here. If you feel the technical details are important, they could perhaps be moved (and expanded upon for clarity) to a text box, but I do not think this is necessary. ok]** Use of OMT **[Au: Correct? Otherwise, the link to OMT was not clear]** in a model of gadolinium uptake of gadolinium, injected into the cisterna magna from the CSF, into the perivascular spaces **[Au: Edited wording correct? Sort of]** now accounts for advection and diffusion flow and for image noise when estimating the movement of gadolinium through the image, enabling visualization of fluid movement **[Au: “glymphatic system” replaced with “fluid movement”. OK? Ok but you will have ot mention the g word somewhere]** over several hours³³. The ability to visualize and quantify dynamic perivascular space function in humans, which could be enabled by application of OMT to dynamic intravenous-gadolinium-enhanced MRI⁶⁶, could also be very powerful ~~when available~~. **[Au: You have added “when available” here – is it likely to be available soon, or is this yet to be developed? Please expand to make clear the current status of this technology. IT IS A MATHEMATICAL ANALYSIS PROCESS WHICH HAS NOT BEEN TRIED YET BUT HAS A REASONABLE CHANCE OF WORKING GIVEN THAT OMT WORKS ON OTHER SYSTEMS WITH SIMILAR CONSTRAINTS]**

[H2] Insights from MRI **[Au: Heading shortened to fit our character limits. OK? OK]**

We do not yet know precisely why perivascular spaces in humans become visible on MRI or post-mortem. However, we do know that several factors are associated with this increase in visibility, and these factors could indicate underlying mechanisms and neurological implications. Below, we discuss the major associations that have been identified; other risk factors have been reported, but our focus is on those for which the evidence is strongest. [Au: In your rebuttal, you made clear that you've focused on the major associations with strong evidence, but I felt this could be clearer in the manuscript itself. Addition OK? OK]

[Au: The following paragraph was no longer appropriate at the end of the section because it applies to all of the subsections that were created, so I have suggested moving it to here to provide an overall summary before the details are given in the subsections below. OK? OK] In combination, the results of the studies show that the associations of visible perivascular spaces differ according to perivascular space location. Hypertension, systemic markers of inflammation, lacunar stroke and dementia are more strongly associated with visible perivascular spaces in the basal ganglia than with perivascular spaces at other sites. These differences might reflect differences in the anatomy or function of perivascular spaces, although some caution is required in the interpretation of the findings in view of the substantial variation in study methods, populations and co-variate adjustment¹.

[H3] Risk factor associations

Visibility of perivascular spaces increases with age. Strong evidence for this association comes from a risk-factor-adjusted meta-analysis of 13 studies that included a total of 8,395 individuals¹. [Au: Edited sentence OK? Ok Reference citation OK? ok] Perivascular space visibility was assessed with visual scores, which increased with age in the basal ganglia, centrum semiovale and hippocampus¹. The age–visibility association differed between the three areas ($\chi^2 = 7.1$, $P = 0.03$) and was strongest in the basal ganglia (OR 1.47, 95% CI 1.28–1.69, $P < 0.00001$), weaker in the centrum semiovale (OR 1.26, 95% CI 1.07–1.49, $P = 0.005$), and weakest in the hippocampus (OR 1.14, 95% CI 1.01–1.30, $P = 0.03$).

Perivascular space visibility also increases with hypertension on the basis of 11 studies, including 7,872 individuals who were mostly treated for their hypertension [Au: OK? ok]¹. [Au: Edited sentence and citation of reference 1 OK? yes] This association was significant in the basal ganglia (OR 1.67, 95% CI 1.20–2.31, $P = 0.002$) but not in the centrum semiovale (OR 1.42, 95% CI 0.92–2.20, $P = 0.12$)¹. However, perivascular space visibility was not associated with diabetes mellitus (five studies, 3,095 individuals)¹. Data were insufficient for a meta-analysis of the

association between visibility of perivascular spaces and smoking, but results of one study indicate no association⁶⁴.

An association between perivascular space enlargement and inflammation must also be considered because histopathological studies show that perivascular spaces are sites of inflammatory infiltrates in SVD. [Au: Please cite appropriate reference(s) to support this statement refs 35, 36.] In MS — an inflammatory disease — focal perivascular space widening has been observed at the edges of active MS plaques at the start of inflammatory exacerbations^{10,61}. In one report, these prominent focal perivascular spaces disappeared as the lesion inflammation resolved¹⁰. Furthermore, evidence suggests that the visibility of perivascular spaces in the basal ganglia and the centrum semiovale of patients with systemic inflammation (systemic lupus erythematosus (SLE)) is higher than that in healthy age-matched controls^{67,68} but of similar severity to that in age-matched patients with minor stroke, despite the much lower vascular risk factor profiles of patients with SLE⁶⁸. Patients with SLE also had more WMHs than expected for their age, and these features and other microstructural brain damage were associated with minor cognitive impairments⁶⁹ and symptoms of fatigue⁷⁰. Similarly, in a community-based study of ageing that included ~700 individuals all aged 72 years, higher plasma levels of inflammatory markers were associated with higher visibility of perivascular spaces in the basal ganglia, which in turn was associated with a greater number of WMHs, but plasma levels of inflammatory markers and the number of WMHs were not directly associated⁷¹. This observation suggests that the association between inflammation and WMHs seen in other studies is secondary to inflammation-related dysfunction of perivascular spaces, but confirmation of this hypothesis requires further investigation.

[H3] Disease associations [Au: Change to “Disease” OK? ok]

Visible perivascular spaces have been associated with a range of neurological conditions. However, small numbers of studies, variation in study populations and variations in clinical phenotypes have ~~prevented~~impeded meta-analyses, meaning the evidence is unclear. [Au: OK? “sketchy” seemed a bit colloquial ok] The available evidence in various conditions is discussed below. [Au: Addition made to lead into discussion. OK? ok]

In seven cross-sectional studies involving patients with various types of stroke ($n = 2,855$), no overall associations have been found between perivascular spaces and stroke subtypes, [Au: Is the meaning here that no association is seen when data from all seven studies are combined, or that no associations have been

found in any one of the seven studies? The original wording suggested the former (and I have edited to retain that meaning), but from what follows, I understand that the latter is the case. Please clarify - the former interpretation is correct] although suboptimal subtyping (of, for example, lacunar versus non-lacunar stroke) in some studies¹ might have masked an effect — individual studies in which subtyping was optimal did show that basal ganglia perivascular spaces were more prevalent in lacunar stroke than in non-lacunar stroke⁵⁵. The picture is also mixed for recurrent stroke⁵. In a study of 2,002 patients with ischaemic stroke or transient ischaemic attack (TIA), recurrent ischaemic stroke (but not recurrent haemorrhagic stroke) was more common among patients with a larger number of visible perivascular spaces [Au: OK? ok] than among those with fewer visible perivascular spaces⁵. [Au: Sentence broken here for readability. Citation of reference 5 OK?OK] In a study of 229 patients with intracerebral haemorrhage and amyloid angiopathy, high visibility of perivascular spaces in the centrum semiovale was associated with recurrent intracerebral haemorrhage⁵. However, in a study of 1,228 community-dwelling individuals, high perivascular space visibility was associated with higher risk of any vascular death but not of incident stroke or event of and death but, overall after adjustment for vascular risk factors, was not associated with recurrent stroke⁵. [Au: Each of the three last sentences seems to refer to a single study, yet the same meta-analysis seems to be cited for all three. Given that they are single studies, I think it would be preferable to cite the studies described rather than the meta-analysis.ok]

The MRI data on associations of perivascular spaces with cognitive decline and dementia are conflicting. An early study suggested that increased visibility of perivascular spaces in the basal ganglia and hippocampus was associated with cognitive impairment in patients with insulin-dependent diabetes and hyperglycaemic episodes⁴⁴. In three other studies ($n = 1272$), no association was found between visibility of perivascular spaces in the basal ganglia and cognitive impairment. [Au: Please cite the three studies here.] One study indicated a borderline significant association of perivascular space visibility in the centrum semiovale with cognitive impairment, [Au: Please cite this study here] three studies have shown that visibility of perivascular spaces was greater in patients with dementia than in controls, [Au: Please cite these three studies here] and two studies have shown that increased visibility of perivascular spaces is associated with cognitive impairment^{1,3}. [Au: Please cite the specific two studies for the final part of the sentence at the end of the sentence.]

Kommenterede [WJ2]: Ref Lau KK, Li L, Lovelock CE, et al. Clinical correlates, ethnic differences, and prognostic implications of perivascular spaces in transient ischemic attack and ischemic stroke. *Stroke*. 2017; 48(6):1470-1477. doi:10.1161/STROKEAHA.117.016694

Kommenterede [WJ3]: Boulouis G, Charidimou A, Pasi M, et al. Hemorrhage recurrence risk factors in cerebral amyloid angiopathy: comparative analysis of the overall small vessel disease severity score versus individual neuroimaging markers. *J Neurol Sci*. 2017; 380:64-67. doi:10.1016/j.jns.2017.07.015

Kommenterede [WJ4]: Gutierrez J, Elkind MSV, Dong C, et al. Brain perivascular spaces as biomarkers of vascular risk: results from the Northern Manhattan Study. *AJNR Am J Neuroradiol*. 2017;38(5):862-867. doi:10.3174/ajnr.A5129

Kommenterede [WJ5]: Uiterwijk R, Huijts M, Staals J, et al. Subjective cognitive failures in patients with hypertension are related to cognitive performance and cerebral microbleeds. *Hypertension* 2014; 64: 653–657.

Kommenterede [WJ6]: Zhu YC, Tzourio C, Soumare' A, Mazoyer B, Dufouil C and Chabriat H. Severity of dilated virchow-robin spaces is associated with age, blood pressure, and MRI markers of small vessel disease: a population-based study. *Stroke* 2010; 41: 2483–2490. Patankar TF, Mitra D, Varma A, Snowden J, Neary D and Jackson A. Dilatation of the Virchow-Robin space is a sensitive indicator of cerebral microvascular disease: study in elderly patients with dementia. *AJNR Am J Neuroradiol* 2005; 26: 1512–20. Beak HW, Choi HJ, Byun MS, et al. Prevalence of enlarged perivascular spaces in a memory clinic population. *Alzheimers Dement* 2015; 11: P146.

Kommenterede [WJ7]: Chen W, Song X and Zhang Y. Assessment of the Virchow-Robin Spaces in Alzheimer disease, mild cognitive impairment, and normal aging, using high-field MR imaging. *AJNR Am J Neuroradiol* 2011; 32: 1490–1495. MacLulich AMJ, Wardlaw JM, Ferguson KJ, Starr JM, Seckl JR and Deary IJ. Enlarged perivascular spaces are associated with cognitive function in healthy elderly men. *J Neurol Neurosurg Psychiatry* 2004; 75: 1519–1523.

With respect to future risk of dementia, one population-based study ($n = 1,778$) showed that the ~~top grade~~ highest score [Au: Please clarify what you mean by “top grade” in this context] of visible perivascular spaces in the basal ganglia and centrum semiovale predicted incident dementia. [Au: Please cite this specific study here] Similarly, another population study ($n = 2,612$) showed that large perivascular spaces overall and particularly visible [Au: OK? Is now] perivascular spaces in the basal ganglia predicted a steeper decline in processing speed and were associated with an increased risk of incident vascular dementia, but not of all dementia or AD⁵. [Au: Please cite the specific study here. 72] ⁷² [Au: Citation of reference 72 was not deleted, but the sentence it was at the end of was deleted in your revision. Please clarify whether this citation should be at the end of the previous sentence or should be removed from here. Retain - see above] In five population-based cohorts including a total of 3,575 individuals with a mean age of 63 years, no association was seen between visual scores of perivascular spaces and current cognitive status⁷³. However, the participants might have been too young or too healthy for an association to be apparent, or visual scores might have been too insensitive to detect subtle associations with cognition.

Studies of perivascular spaces in other neurological disorders suggest that visible perivascular spaces on MRI [Au: OK?ok] are associated with autism spectrum disorders, depression, Parkinson disease, neurolupus, MS, CADASIL⁵⁴, collagen type IV mutations⁷⁴ [Au: Please cite studies for autism spectrum disorders, depression, Parkinson disease, neurolupus and MS; these are all summarised in ref 1 – it would be a lot of extra refs to cite individually] and a range of other conditions, detailed in reference 1. However, these studies have mostly been small and many did not include adjustment for vascular risk factors¹, meaning these associations require further investigation and confirmation. [Au: Addition made to make implication clear. OK? ok] Finally, the relationship between perivascular space visibility and mortality is also unclear. In one study of patients with ischaemic stroke ($n = 2,002$), no association was found between perivascular spaces and death. [Au: Please cite this specific study] By contrast, a population study ($n = 1,228$) showed that visibility of [Au: OK?] PVS was associated with death from vascular causes⁵. [Au: Please cite the specific study rather than the meta-analysis]

[H3] Association with small vessel disease lesions [Au: Change of heading OK to avoid abbreviations? ok]

Kommenterede [WJ8]: Zhu YC, Dufouil C, Soumare A, Mazoyer B, Chabriat H, Tzourio C. High degree of dilated Virchow-Robin spaces on MRI is associated with increased risk of dementia. *J Alzheimers Dis.* 2010; 22(2):663-672. doi:10.3233/JAD-2010-100378

Kommenterede [WJ9]: This is ref 72

Kommenterede [WJ10]: Actually most of this is already covered on the prior page and I think could be deleted. It's the same few studies as already cited.

Use of MRI to investigate perivascular spaces in humans began in earnest [Au: Changed to “began in earnest” because it didn’t seem logical to say that this and research into microbleeds have both “come to prominence” but for microbleeds to have received considerably more attention. Change OK? ok] in the mid-2000s⁴⁶, several decades after WMHs⁷⁶ and lacunes^{77,78} had been established as clinical features of SVD [Au: Clarification of the way in which these had become established added – OK? ok] and shortly after microbleeds came to prominence⁷⁹. However, microbleeds have received far more attention than perivascular spaces owing to concerns about their association with the risk of haemorrhage with antithrombotic or thrombolytic treatments.

Early studies from this period suggested that perivascular spaces — at least those in the basal ganglia — were associated with WMHs and lacunes in patients with stroke or cognitive presentations of SVD^{46,55} and in community-dwelling older [Au: Can you be more specific about the meaning of “older” here – above a certain age? Generally means people in their 60s, 70s or above] people^{64,80}. However, potential overlap in size between perivascular spaces and small lacunes [Au: Edited wording OK? ok] led to suggestions that perivascular spaces might be mistaken for lacunes, thereby causing overestimation of the association. In a study published in 2013, analysis of reported sizes of perivascular spaces and lacunes in the literature showed that, although the largest reported perivascular spaces were 5 mm in diameter, most had a maximum diameter of 3 mm, whereas most lacunes had a minimum diameter of 3 mm (refs⁸¹), suggesting minimal overlap. [Au: Interpretation added to spell it out. OK? ok] However, the sensitivity of MRI has increased since the 2000s, so these cut-offs might need revision. In addition, we have observed [Au: This wording implies that this is unpublished data – correct? yes] that small subcortical infarcts with diameters of up to 10 mm when acute can resolve to leave a lacune with a diameter <2 mm, so size criteria for perivascular spaces should be seen as a guide and applied with caution pending further data from longitudinal studies of the fate of small subcortical infarcts. The variation in the sizes of perivascular spaces, and the overlap with the ~~maximum minimum~~ diameter of lacunes, ~~are consistent with pathological findings~~ [Au: Which pathological findings? Please clarify – suggest remove statement to simplify] ~~and~~ might explain why lacunes are most commonly ~~reported to be seen~~ in the basal ganglia — ~~if~~ at least some lacunes might actually be perivascular spaces⁸². [Au: It wasn’t clear in what way this might explain the preponderance of lacunes – I have edited according to my understanding, but if this is incorrect, please change and clarify the logic of the original statement. Hope now clearer]

Many cross-sectional studies have demonstrated that perivascular spaces are associated with WMHs — the more visible perivascular spaces, the more severe the WMHs, and WMHs seem to form around perivascular spaces (Figure 3). [Au: Please cite appropriate references to support this sentence its an observation – use ref in comment box and 72] However, a systematic review of associations between perivascular spaces and SVD lesions published in 2019 questions whether perivascular spaces are cross-sectionally associated with WMHs¹. Across the 23 studies included in this analysis and in which adjustments were made for age and vascular risk factors, perivascular spaces were associated with lacunes ($n = 4,894$, OR 3.56, 85% CI 1.39–9.14, $P = 0.008$) and microbleeds ($n = 5,015$, OR 2.26, 95% CI 1.04–4.90, $P = 0.04$), but for WMHs, the results were consistent with a positive association but were not significant ($n = 4,974$, OR 1.54, 95% CI 0.71–3.32, $P = 0.27$). Of another eight studies that could not be included in the meta-analysis, seven indicated a positive association between basal ganglia perivascular spaces and WMHs¹. Therefore, the lack of an association in the meta-analysis might have resulted from methodological limitations. Few longitudinal studies of perivascular spaces and WMH development have been conducted. In one population-based study, large perivascular spaces on MRI were associated with worsening of WMHs and increasing lacune formation over 4 years; WMH progression was more severe with a greater number of large perivascular spaces⁷².

Kommenterede [WJ11]: Wardlaw JM, Smith C, Dichgans M. Small vessel disease: mechanisms and clinical implications. *Lancet Neurol* 2019; 18:684-696

Investigation of microbleeds and perivascular spaces has revealed associations between microbleeds and basal ganglia perivascular spaces (five studies, $n = 5,015$, OR 2.26, 95% CI 1.25–4.00, $P = 0.04$) and between microbleeds and centrum semiovale perivascular spaces (two studies, $n = 367$ [Au: Please add n for consistency] OR 1.15, 0.62–2.12, $P = 0.66$); the available data are insufficient to determine whether the association differs between lobar and deep microbleeds. [Au: Edited sentence OK? ok] In a longitudinal study, a high number of large perivascular spaces was associated with worsening of microbleeds over 4 years⁷². In superficial siderosis, which can be — another manifestation of cerebral amyloid angiopathy (CAA) in addition to microbleeds [Au: No other manifestation of CAA has been specifically mentioned – please clarify – microbleeds are a manifestation of CAA. Please also clarify how this passage on superficial siderosis follows on from the discussion of microbleeds – amended to clarify] — focally prominent perivascular spaces can occur beneath affected areas of cortex⁸³, supporting the idea that perivascular space dilation is a sign of blocked drainage pathways as a result of A β deposition¹⁵, which is indicated by concurs with post-mortem evidence (Figure 3)⁵⁶. [Au: Edited sentence OK? Indicated by was a bit strong] However,

Formateret: Skrifttype: Kursiv

perivascular spaces become visible in the white matter up to the inner cortical margin in many non-CAA-related contexts (Figure 2), indicating that this feature cannot be specifically related to CAA or A β . Instead, perivascular space dilation in the immediate subcortical white matter could indicate impaired fluid drainage, particulate or protein deposition, excess fluid leakage from the vasculature^{84,85} and/or stagnation for any reason; the concept of protein elimination failure angiopathies (PEFA) has been proposed to describe these clinical phenomena^{8,27,58}. [Au: Edited sentence OK?ok] [Au: The final point in this paragraph about the original isolation of A β didn't seem necessary]

Modelling has shown that the four commonly observed MRI features of SVD (WMHs, lacunes, microbleeds and perivascular spaces) can be used to derive a 'total SVD burden' latent variable, a measure that is similar to a summed SVD burden score and that is similarly associated with general cognitive ability in old age⁸⁰. [Au: Edited to streamline and simplify. OK?ok] All four variables contributed to the latent SVD variable. [Au: The detail of the magnitudes did not seem necessary to making the main point here, so I have suggested they are removed to simplify this passage. OK?ok] Therefore, besides the evidence for associations of PVS with WMHs and microbleeds, it is probably reasonable to work on the principle that perivascular spaces are associated with all SVD lesions. [Au: Changes to sentence OK?ok] However, evidence for a causal role of perivascular spaces in SVD lesion formation is very limited, and perivascular spaces that are visible on MRI could be an epiphenomenon or could represent an early stage in the development of other SVD lesions.

[H1] Anatomy, physiology and fluid drainage [Au: Change of heading OK?ok] [Au: Since there is such a notable transition of topics at the start of this section, I feel that a brief introduction is needed to this section to provide the reader with an overview of why these aspects are being discussed. I have suggested a draft version of this, but please feel free to edit this or write an alternative if you feel that what follows can be reflected more accurately or in a different way. Thank you. The draft is fine] Evidence indicates that perivascular spaces have important roles in the movement and drainage of fluid in the brain and that dysfunction of perivascular spaces is associated with neurological disorders. However, the anatomy of perivascular spaces is complicated and controversial (Box 1), and gaps in our knowledge mean that their precise roles in fluid movement and drainage in humans are unclear, hindering clinical translation. In this section, we discuss the anatomy and physiology of perivascular spaces and their involvement in

fluid movement and drainage in the brain, with the aim of highlighting gaps in our knowledge and the research needed to fill these gaps.

[H2] Perivascular space anatomy

In mammals, the perforating arterioles, which run deep into the brain, are surrounded by pia mater at the surface and within the brain (Figure 5)^{8,30}. [Au: Change to wording OK?ok] At the capillaries, the pial membrane is replaced by a basal lamina that is surrounded by a 'cuff' of astrocyte end feet^{30,40}. [Au: Citation of Figure 6 was not appropriate hereok] The space between the arteriolar smooth muscle basement membrane and the enveloping pia mater is generally referred to as the Virchow–Robin space and corresponds with the perivascular spaces that are visible on MRI.

Arterioles that perforate through the base of the brain to enter the basal ganglia [Au: Are these arterioles distinct from the perforating arterioles discussed above, or are they a subset? Please clarify they are a subset that enter via the basal ganglia as opposed to through the cortical surfaces] are ~~surrounded-invested~~ by two layers of leptomeninges along the arteriole length [Au: Are they surrounded by two layers along their length, or just once they enter the basal ganglia? Please clarify] and the perivascular spaces are between these layers (Figure 5). In rodents and humans, these perivascular spaces communicate with the basal subarachnoid space^{4,26}. Arterioles that enter the brain via the convexity cortex, and all venules, are surrounded by one leptomeningeal layer⁵⁶. [Au: Are these “arterioles that enter the brain via the cortex” different from the perforating arterioles discussed above? If so, how? Different in their anatomical location and also in their individual perivascular anatomy which is being explained here. Also, do the arterioles that enter the basal ganglia also not initially enter the brain through the cortex? No Please clarify the distinctions between the different groups of arterioles mentioned. Hope now clearer, and also relevance to earlier statements about different risk factor and disease associations with PVS in different regions]

Whether periarteriolar spaces around arterioles perforating through the convexity cortex communicate with the subarachnoid space or only with the subpial space is currently under debate (see earlier 'Anatomy' and ~~below the following~~ [Au: Please specify the relevant sections])^{26,82,87,88}. Histological studies of rodents suggested that these periarteriolar spaces only reached the subpial space^{26,82,87,88}, but these studies were conducted with post-mortem tissue. In subsequent studies in live mice, perfusion fixation resulted in abnormal retrograde flow in the perivascular

spaces, collapse of the perivascular spaces and deposition [Au: OK?ok] of tracer in the vessel wall^{13,89} hence tracer destination may be influenced by artefact. [Au: Please explain the implication of these findings. How does it relate to the debate about communication with the subarachnoid/subpial space? Please see my comments on the figure below – I didn't feel that this point justified a figure.]

[Au: From comments of the reviewers and in your rebuttal, I had understood that perivascular spaces are not present in the cortex, so I am a little confused by discussion of "cortical perivascular spaces". Does this refer to perivascular spaces around vessels that enter via the cortex? Or am I misunderstanding the nomenclature? If you are able to clarify this apparent conflict, that would be extremely helpful.] Cortical perivascular spaces must communicate with the subarachnoid CSF somewhere because tracer that is administered to live rodents via the cisterna magna CSF is rapidly taken up into perivascular spaces in the convexity cortex and in the basal ganglia⁹⁰. Similarly, in humans, gadolinium (an MRI contrast agent) injected into lumbar CSF spread to the perivascular spaces in the [Au: Addition OK? ok Otherwise, I don't think it's clear how these findings tell us that the perivascular spaces communicate with the CSF, so if this change is not correct, please spell out the interpretation of the observation] convexity cortex of all brain regions⁹¹ by 6–9 h after injection and had spread to perivascular spaces [Au: OK? See previous commentok] throughout the cortex and white matter by 24 h (Figure 6)⁹², a similar distribution to that in rodents. In contrast to rodents, however, little gadolinium uptake was seen in perivascular spaces in the basal ganglia in humans, [Au: Sentence edited to emphasize the difference from rodents. OK?ok] despite the basal ganglia being a prominent site of perivascular space visibility on MRI in humans¹, and despite the existence of [Au: Change to wording OK? ok "a prominent site of...open connections" did not seem to make sense] open connections between the basal cisterns and the basal ganglia perivascular spaces (Figure 2)⁴⁸. In agreement with this finding, in patients undergoing investigations for normal pressure hydrocephalus, gadolinium uptake into basal ganglia perivascular spaces was lower than that in controls (patients undergoing investigation for a suspected CSF leak)⁹³. Patients also showed more gadolinium reflux into the lateral ventricles and periventricular white matter and delayed clearance of gadolinium. [Au: I have removed the additional findings of this study, because they seemed to distract from the main point being made about the uptake in the basal ganglia. OK? No these were important points demonstrating failure of Gd clearance – main points reinstated] Differences in

Kommenterede [WJ12]: Perivascular spaces are present in the cortex but are the topic of debate since they are less obvious than in other tissues. The solution to this conundrum is not yet clear.

the uptake of tracers into perivascular spaces between studies and between animals and humans might reflect differences in positioning during imaging. [Au: Addition made to provide the general issue before the details are explained. OK?ok] Rodents are typically in the prone (ventral side down) position for 2-photon imaging and in the supine (dorsal side down) position for MRI⁴⁸. Humans typically undergo MRI in the supine position. Gadolinium is heavier than CSF, so gravitates to lower locations, and body position substantially affects the distribution of CSF tracers and contrast agents in rats⁹⁵. Nevertheless, the conflicting data that are currently available leave the precise anatomy of perivascular spaces unresolved, and further work is needed to address the uncertainties. [Au: Sentence added to emphasize the fact that there are unknowns, in line with the aim of the review that has been clarified in the introduction. OK? perfect]

[H2] Fluid drainage pathways [Au: Heading OK?ok]

[Au: The initial passage about CSF, ISF and their drainage below is quite confusing for the following reasons:

- Unlike for CSF, there is no clear introduction of what interstitial fluid is, where it originates from and how it relates to CSF. From my understanding, it is produced at the blood brain barrier and moves to into the CSF, with which it drains via the same pathways. I would suggest adding an explanation of this (or the correct version if this is incorrect!), as it is important for the reader to understand the route that fluid is taking through the brain.
- CSF and ISF seem to be used almost interchangeably when discussing drainage, which makes it unclear whether they drain via different pathways or the same pathways. Please ensure that it's clear whether drainage of interstitial fluid and of CSF are equivalent/occurs via the same pathways.]

Humans have ~140 ml of CSF, which is produced continuously in the choroid plexus and passes through the ventricles to the subarachnoid space. From here, CSF is thought to drain into the sagittal and transverse venous sinuses via arachnoid granulations (structures that project from the subarachnoid space through the dura mater into the sinuses), [Au: Addition made to more clearly explain the anatomy. OK?ok If incorrect, please amend] although the extent to which arachnoid granulations are involved in drainage of CSF has been questioned⁹³. [Au: Edited version OK?ok] In humans, the dural venous sinuses also have many arachnoid

villi, which are smaller than granulations. –[Au: Are arachnoid villi different to arachnoid granulations? Please explain] and that are not visible on MRI but can mediate drainage of CSF. [Au: Addition of “but can mediate drainage of CSF” OK? Please cite reference(s) to support this statement.] In rodents, arachnoid granulations are mainly present in ventral venous sinuses, and the extent of their involvement in CSF drainage is also unclear. Although still under debate, other likely sources of fluid that ultimately enters the CSF ~~production~~ include ~~the generation of~~ ISF formed from fluid crossing the ~~from the cerebral microcirculation via the~~ BBB²⁵ from the cerebral microcirculation and as a byproduct of brain metabolic activity. [Au: Is this sentence necessary? **YES** Would it be sufficient to just say above that most of the CSF is produced in the choroid plexus. **No** If you feel this sentence is necessary, please clarify the following aspects of the meaning. Do you mean to say that the ISF becomes incorporated into CSF? **Yes** What do you mean by “via the BBB” – the wording implies “generation...via the BBB”, the meaning of which is not clear. I have clarified Finally, do you mean that ISF is generated as a byproduct of brain metabolic activity? **Yes. Water is a byproduct of energy metabolism** Please also cite a reference to support the statement about the byproduct of metabolic activity.]

[Au: While CSF is introduced clearly above, ISF is not – I think it would be helpful here to introduce what ISF is, where it originates, how it relates to CSF, and therefore how it drains from the brain. This will help the subsequent discussion of drainage routes make sense to readers who are not familiar with this field.] Some ISF might return to the circulation by exchange across the BBB and resorption into capillary-venular blood, and therefore drain out with the blood. However, substantial evidence indicates that perivascular spaces are important in the drainage of ISF, although the precise pathways and mechanisms remain unclear. [Au: Sentence added to create a transition into the discussion of PVS involvement. OK?**ok**]

Many previously published diagrams in review articles suggest that most ISF leaves the brain via the perivenular spaces^{29,90,95,96}, but the evidence in support of this hypothesis is limited ~~and unclear~~. [Au: Addition to end of sentence OK?] Whether the perivenular spaces connect to subarachnoid CSF, which would allow drainage, [Au: Addition OK?] or remain subpial, which would ~~not then drain by a different route~~, [Au: Addition of “which would allow drainage” and “which would not” correct? If not, please clarify what the implications of the two alternatives are] remains uncertain^{8,26,17,27,29,40,53,97}. Furthermore, most primary studies of ISF drainage [Au: “of ISF drainage” correct?**ok**] have focused on periarteriolar spaces,

Kommenterede [WJ13]: All tissues anywhere in the body have interstitial fluid. It is formed from fluid that enters the tissue from the microcirculation, and in highly metabolically active organs water (and Co2) is also a byproduct of metabolism. ATP breaks down in the presence of O2 to CO2 and H2O. Failure to drain ISF or overproduction through eg leaky blood vessels, results in tissue oedema. This is general physiology.

so data on perivenular spaces are limited. In one study published in 2013, lipophilic and hydrophilic fluorescent tracers were present in the perivenular space at 60–90 mins after cisternal injection in mice⁹⁸, indicating that these spaces are a channel of ISF drainage. [Au: Addition of interpretation OK for clarity? If not, please correct interpretation ok] However, few other published images — perhaps only one to date⁹⁰ — provide convincing evidence that contrast agent reaches perivenular spaces by any route. Other studies have shown [Au: The meaning of “differentiated between ISF drainage and CSF uptake” was not clear and this phrase didn’t seem necessary, so I have suggested it is removed. OK?] that CSF tracers injected into the cisterns [Au: OK?ok] entered the brain along pial–glial membranes and left the brain along capillary basement membranes and arteriolar smooth muscle cell basement membranes against the direction of blood flow^{99,100}. ~~This observation suggests that without involving~~ perivenous spaces ~~are not involved~~ in elimination of ISF and solutes from the brain¹⁰⁰. [Au: Edited sentence to simplify. OK? I have simplified further]

In the context of A β deposition — such as in CAA or AD, in rodents that overexpress A β , or after A β injection into the hippocampus⁹⁹ — A β is deposited in pericapillary and periarteriolar membranes but not in perivenular membranes¹⁷. The periarteriolar and pericapillary localization of vascular amyloidosis might reflect the many contractile cells (which are thought to be phagocytic to A β) around arterioles and capillaries and a sparsity of contractile cells around venules. [Au: I could not see how the fact that smooth muscle cells and pericytes are phagocytic related to the sparsity of contractile cells, so I have removed that part of the sentence. OK? However, please explain more clearly why the periarteriolar and pericapillary localization tells us about the contractile cells around venules, and what the implications are of this sparsity of contractile cells clarified] Tracers injected into the rodent brain parenchyma seem to drain mainly along arterioles, not venules [Au: Please explain what this observation tells us] although striatal injection ~~could be considered~~ is not physiological and the pressure could force tracer along lines of least resistance that are not necessarily physiological paths—pathological¹⁰¹, similar to seeing fresh intrastratial haemorrhage tracking in basal ganglia perivascular spaces in humans (JMW, personal observation). [Au: The meaning of this sentence is not clear. How does striatal injection being pathological relate to the tracers draining along arterioles? And how is this “similar to seeing fresh haemorrhage tracking...”? This statement assumes the reader knows what this is, but it is not mentioned previously. Hope now clearer]

[Au: I have suggested the following paragraph to emphasize that the evidence indicates that perivenular spaces are not involved but that there might be experimental reasons for this, and that more research is therefore needed, thereby highlighting more clearly the knowns and unknowns. OK?]

Much of the available evidence, therefore, suggests that periaarteriolar spaces rather than perivenular spaces are involved in drainage of ISF from the brain. However, the ~~near-total-absence~~lack of evidence that fluid reaches perivenular spaces might be a consequence of experimental limitations, such as short durations of experiments focused on periaarteriolar spaces not allowing enough time for tracer to pass through the interstitial space and far enough into perivenular spaces in sufficient quantities to be detected. [Au: Why would the short durations have prevented observation of tracer in perivenular spaces? Does it take longer for tracer to reach perivenular spaces? If so, why?] It might be due to the dilution of tracers as they pass through CSF and ISF reducing the tracer's visibility. [Au: Do you mean that tracers are diluted too much before administration, or that they become diluted in the system? Please clarify] Or it might be differences in fluid flows between physiological and pathological states, or fixation artefacts. Further research is therefore needed to resolve this question.

[H3] *Meningeal and nasal lymphatics*

[Au: In this section, I have cut down the detail on the lymphatics themselves and the experimental details related to them, and I think that instead some discussion of how they relate to perivascular spaces is necessary to justify the inclusion of this passage in the article. Currently, there is no clearly explained link between the meningeal and nasal lymphatics and the perivascular spaces. Are these lymphatics a potential destination for fluid being moved along perivascular spaces? If so, I think this needs to be made clear. If the lymphatics represent a drainage system that is separate to perivascular spaces, I think it only warrants a much more brief mention as an alternative pathway and a brief discussion of the fact that the contribution of the different pathways to drainage is unknown.]

Meningeal lymphatics drain alongside the major dural venous sinuses, via cranial perineuronal channels^{26,99}, to cervical lymph nodes^{8,105}, and are thought to be major drainage routes for interstitial fluid^{105,106}. [Au: Statement about the history removed. OK?ok] Probable small dural lymphatics have also been visualized in humans in vivo^{28,107}. In mice, dural lymphatics have been identified that run along the major venous sinuses¹⁰⁸. [Au: I have cut down this paragraph to give the basic

facts of the anatomy without experimental detail.ok] Rodents, rabbits, pigs, sheep, monkeys and humans also have nasal lymphatics that drain from the frontal subarachnoid space, through the cribriform plate via the nasal mucosa and to the cervical lymph nodes^{8,26,109}.

In vivo tracer experiments in rodents and humans have shown that ISF drains from the brain via nasal lymphatics to the cervical lymph nodes^{26,91}. In addition, PET imaging in patients with AD and healthy volunteers has been used to show that intravenously injected tau ~~can~~ may exit the brain via this drainage route¹¹⁰. although f- Further research is required to determine whether intrathecally administered radiotracers can be used to determine where tau, A β and other disease-associated molecules exit the cranial cavity and which routes are the most important in humans.

The relative proportions of ISF that drain through the two pathways — either via nasal, peri-arterial or peridural lymphatics to the blood or cervical lymph nodes, or via arachnoid granulations and villi into the major venous sinuses — and how the perivascular spaces feed into these exit channels, is unclear. **[Au: Related to my queries above, do these pathways represent one that involves perivascular spaces and one that does not? If so, please make this clear. If perivascular spaces are involved in both pathways, please make clear their importance in both pathways. Its not clear]** In transgenic mice bred to lack~~without~~ dural lymphatics, clearance of macromolecules from the brain was attenuated and fluid drainage to cervical nodes was diminished, but the effects on brain ISF drainage and intracranial pressure were minimal, suggesting that most fluid exits the brain by alternative routes¹⁰⁶. primarily perivascular spaces and thence to venous sinuses rather than via lymphatics. **[Au: Addition OK to explain implications? Could this be more specific and read “suggesting that most fluid exits via the venous sinuses (and therefore the perivascular spaces?) rather than the lymphatics”?** **yes]** Some gaps in our understanding of these processes might be explained by inter-species differences, as most studies of CSF and ISF have been done with rodents, rabbits or sheep⁸.

[H2] Perivascular spaces and blood–brain barrier leakage [Au: I don't think this section sits entirely comfortably in the article because it discusses a very specific aspect that is not really related to the drainage systems discussed above. Please consider whether it is necessary and, if so, whether it could be reduced in length.]

In one study of patients with lacunar or cortical stroke, **[Au: Addition OK?yes]** intravenous administration of gadolinium led to visibility of perivascular spaces in the

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basal ganglia after just 30 mins⁸⁴. The contrast agent could only have reached the perivascular spaces by crossing the BBB somewhere⁸⁴, [Au: See my comment on the figure – I didn't feel that this point justified a figure ok] but how the gadolinium entered the perivascular spaces was unclear because its pathway could not be tracked in real time. One possibility is that the gadolinium crossed into the CSF at the choroid plexus, travelled out of the ventricles and flushed back into the basal perivascular spaces, but this pathway seems unlikely in only 30 mins⁹². Another possibility is that gadolinium crossed the BBB and collected in the basal perivascular spaces, possibly after also entering the ISF from the ISF, [Au: It's not clear how the gadolinium would have got to the interstitial fluid. Please clarify] although the direction of flow appears counter to that observed for tracers injected into cisternal CSF which move up the basal perivascular spaces in rodents and humans. [Au: Firstly, please cite the studies that have shown this. Refs 12, 13, 14, 48, 49, 91, 92, 93, 94 Secondly, please explain why this observation argue against the possibility that Gd collects in PVS from the ISF. Hope now clearer] The most likely possibility is that the intravenously injected gadolinium leaked directly [Au: Addition of "directly" OK?] from the arterioles and/or capillaries into the perivascular spaces [ref 92].

A study published in 2018 showed that this type of direct leakage occurs in pericyte-deficient mice¹⁰². [Au: Sentence edited to tell the reader about this study rather than assuming that they know about it already. OK? ok] These mice developed substantial, early BBB breakdown that led to a large increase in the number of enlarged perivascular spaces in white matter, suggesting that pericyte degeneration that causes BBB breakdown results in increased fluid in (and therefore visibility of) the perivascular spaces¹⁰². Pericyte loss and BBB breakdown occur in neurological disorders that have been associated with visible perivascular spaces in humans¹ and rodents, including AD^{85,103} and CADASIL¹⁰⁴. Therefore, this evidence provides a possible [Au: Addition of "possible" OK? ok] mechanistic explanation for the association between BBB breakdown and perivascular space visibility⁶⁶, but does not help to solve the conundrum of ISF drainage routes (Box 1).

[H2] Drivers of fluid movement [Au: Heading shortened to fit our character limits. OK? ok]

[Au: The introduction of the term 'glymphatic system' seemed to confuse things, and it was not clear how the glymphatic system and perivascular spaces equate/relate to each other. Given that the discussions of mechanisms below refer to perivascular spaces, I suggest removing mention of the

glymphatic system and making this section about movement of fluid in perivascular spaces. I have therefore suggested an introductory paragraph below and then subsections for each of the mechanisms discussed.]

Movement of fluid through perivascular spaces seems to depend on several factors, including vascular pulsation^{13,14,88}, respiratory movement¹¹⁴ and the sleep–wake cycle¹². Some evidence suggests that aquaporin 4 (AQP4) water channels also contribute, ~~but their role is unclear~~. These mechanisms are discussed below. [Au: Paragraph constructed to lead into subsections that discuss each of the mechanisms. OK?ok]

Kommenterede [WJ14]: Prefer to omit – not necessary

[H3] Vascular pulsation

[Au: I have put together all the information on vascular pulsatility into one section. OK?ok] In the periphery, arterial pressure and flow pulsations dissipate to become essentially continuous at the arteriolar and venous level. In the rigid skull, however, limited tissue compliance promotes propagation of arterial pressure pulsation throughout the brain, leading to measurable pulsatile flow in the microvasculature and venous outflow⁵². This preservation of pulsatility along the entire vascular bed is thought to help move fluid and waste through the brain, and might in part explain why periarteriolar (being at the upstream end of the circulation) rather than perivenular spaces (at the downstream end) dilate in disease. [Au: I feel like the logic is not quite complete here, and it's not clear how this might explain why periarteriolar spaces rather than perivenular spaces dilate. Eg., if the pulsatility is preserved along the entire vascular bed, why wouldn't perivenular spaces also be affected? Well the pressure exerts its effect at the proximal end at least to start with] The shape of periarteriolar spaces seems to play a role in efficient fluid shift — the optimal cross-section is oblate (a sort of double-pointed teardrop) with the arteriole lying off-centre^{13,89}. [Au: Is “off-centre” what you meant? Yes If not, please clarify] On this basis, we speculate that minor changes in CSF pressure, tissue pressure or vascular function could alter the shape of perivascular spaces and accelerate dysfunctional flow in perivascular spaces and the development of secondary drainage problems.

In rodent studies, various techniques have demonstrated that movement of fluid along the perivascular spaces is assisted by pulsation in the arteriole at the centre of the space^{13,14,88}. In the most recent of these studies, published in 2018, microparticles that were injected into the cisterna magna could be seen (with 2-photon imaging) moving along perivascular spaces (Figure 7). At normal blood pressures, the arteriolar pulsation and particle movement was smooth and

continuously forwards¹³. During acute blood pressure elevation, which increases arteriole stiffness, the pulsations in distal vessels increased in amplitude, resulting in jerky movement of microparticles and intermittent reversal of the flow, leading to a reduction in net flow in the perivascular spaces and failure to clear debris (see supplementary movies in ref¹³).

Evidence for a similar effect of arteriolar pulsatility on perivascular space function in humans comes from phase contrast MRI studies, which showed that an increase in intracranial vascular pulsatility (particularly venous sinus pulsatility), was associated with an increase in perivascular space visibility^{52,128}. [Au: Change to wording OK?ok] This approach also demonstrated that an increase in intracranial vascular pulsatility was associated with reduction in the volume of CSF craniad and caudad fluctuation with each cardiac cycle [Au: Please clarify what about CSF is fluctuating – the volume, the speed of movement, for example?] at the foramen magnum and, in turn, with increased visibility of [Au: Addition of “visibility of” OK?ok] basal ganglia perivascular spaces, suggesting that a decline in foramen magnum CSF stroke volume ~~results in~~ indicates fluid stagnation and less effective flushing of perivascular spaces¹²⁸. The increased intracranial vascular pulsatility seen with phase contrast MRI was also associated with faster transmission of the vascular pulse wave through the brain and with a greater volume of WMHs⁵². [Au: Addition of “with a greater volume of” OK? ok If not, please clarify what “worse” means in this context] If the association between foramen magnum CSF stroke volume and pulsatility ~~CVR, PI~~ [Au: Please define CVR and PI omit CVR] and perivascular spaces are confirmed in other studies^{52,128}, then phase contrast MRI could be a powerful, non-invasive tool to assess the associations between [Au: Addition OK?ok] intracranial haemodynamics and pathology in humans.

[H3] Respiratory motion

Respiratory motion results in cyclical craniad and caudad movement of CSF at the foramen magnum in humans over and above the fluctuation due to cardiac pulsation: inspiration increases flow of CSF into the ventricular system, and expiration has the opposite effect¹¹⁴. This CSF motion can be seen around the foramen magnum and in epidural veins in the spine¹³⁰. During forced inspiration, as the venous blood is drawn downaway from the head [Au: Down from where? The head?] into the thorax, the CSF moves up into the cranial cavity towards the brain. Through this movement of CSF, respiratory motion might also contribute to the flushing of perivascular spaces. This function could provide another explanation for sighing (beyond re-inflating lung alveoli), although this hypothesis remains to be demonstrated in humans¹³⁰.

[H3] Perivascular spaces and sleep

Studies in rodents have shown that uptake of CSF into perivascular spaces and flushing of the ISF are increased during sleep¹². However, very limited data are available on what the function of perivascular spaces is during healthy sleep and how this function is disrupted in disordered sleep in humans. [Au: Sentence edited to simplify. OK?ok] Given that the human brain is larger than the rodent brain, one might expect that constant replenishment of ISF and removal of waste is required rather than only being a sleep-related process, [Au: “rather than only a sleep-related process” OK? If not, please clarify what “constant replenishment” is in contrast to] but no information is available about this. Nevertheless, some evidence suggests a relationship between perivascular spaces and sleep in humans. [Au: Sentence added to lead into following discussion about evidenceok]

In patients who received an injection of gadolinium into the lumbar CSF to investigate hydrocephalus, uptake of gadolinium into perivascular spaces was greater during the night than during the day, despite patients having their sleep disrupted as a result of repeated scanning^{93,94}. Furthermore, indirect evidence suggests perivascular space dysfunction during disrupted human sleep. For example, increased perivascular space visibility was associated with obstructive sleep apnoea ($n = 170$)⁷⁵ and with reduced sleep efficiency in a small study of patients with cerebrovascular disease ($n = 26$)¹¹. In a population-based study, increased visibility of perivascular spaces in the basal ganglia was associated with reduced sleep efficiency¹³¹. In healthy people, one night of sleep deprivation resulted in increased levels of A β in the thalamus and medial temporal structures¹³². In another study in healthy people, [Au: Addition of “in healthy people” OK? ok] disruption of slow-wave sleep led to higher CSF levels of A β the following day, and worse sleep efficiency in the week preceding a lumbar puncture was associated with increased CSF levels of tau¹³³. [Au: I've broken the sentence here – move of reference #133 to here OK?ok] These findings were interpreted to mean that sleep disruption increased neuronal activity and generated more proteins, but an alternative explanation is that disrupted sleep reduced flushing of perivascular spaces, resulting in less clearance of A β and tau.

To investigate these possible explanations, protein production and clearance of ISF should both be considered. [Au: Addition made to the beginning of the sentence to link it more clearly to above paragraph. Edited sentence OK?ok] In a mouse study published in 2019, increases in levels of tau in hippocampal ISF and in CSF [Au: Change to wording correct?ok] were closely related to neuronal

activity but not to the rate of [Au: Addition of “the rate of” OK? ok If not, please clarify what measure of paravascular clearance was used] paravascular clearance¹³⁴. In humans, CSF levels of tau and A β were increased by sleep deprivation and were related to increased production of these proteins. However, the picture is complex, and in addition to glymphatic-mediated bulk flow, the BBB can facilitate clearance of A β ¹⁷. During wakefulness, 85% of A β is reportedly removed via transport across the BBB and just 15% via passive ISF bulk flow^{136,137}. During sleep, total A β clearance is increased by ~2.2-fold — 60% occurs via BBB transport and 40% occurs via ISF flow^{12,17}. These findings indicate that changes in CSF levels of tau and A β result from multiple factors, including production of proteins and other substances during cell activity and changes in clearance with sleep–wake cycles.

[H3] Aquaporin 4 and astrocytes

Aquaporin 4 is a water channel that is found in astrocyte end feet that wrap around capillaries. [Au: Opening of this passage edited to convey more clearly the uncertainty about the role of AQP4 – as it was, the opening suggested that its role was fairly established, so the statement that followed saying that its role is debated seemed contradictory. Edited version OK? ok] Some evidence suggests that AQP4 facilitates movement of CSF from perivascular spaces into the ISF space and active flushing of ISF⁹⁰, but its role in promoting ISF [Au: OK? This wording suggested that PVS are cleared] clearance is debated²¹. The importance of AQP4 was brought into question by observations that uptake of CSF tracer and the interstitial flow rate of fluorescent dextrans in mice were consistent with diffusion and were unaffected by cardiac arrest or by AQP4 knockout²¹. Other evidence also suggests eds that CSF -periarteriolar-flow into ISF and away from arterioles occurred by diffusion¹¹⁶⁻¹¹⁹. [Au: Edited wording OK? ok] However, use of different strains of AQP4 knockout mice has suggested that AQP4 is necessary for the rapid passage of fluid from perivascular spaces into ISF and through the brain¹²⁰. The conflicting results might reflect differences in technique and/or anaesthetic agents used^{53,101}.

Data on AQP4, vascular function and perivascular spaces in humans are very limited¹²¹. In brain biopsy samples from patients with normal pressure hydrocephalus (mean age 70.8 $\pm$ 8.8 years), levels of AQP4 on astrocyte end feet were lower than those in younger controls undergoing brain surgery for other reasons (mean age 44.0 $\pm$ 16.5 years)¹²², but the age disparity renders these results uninterpretable [Au: I felt that “unreliable” had slightly the wrong connotation — ie. that there was something wrong with the data, rather than that you just can’t draw any conclusions. Change OK? ok] because all known aspects of cerebral

Kommenterede [WJ15]: Yes PVS are cleared by flushing into ISF via AQP4

haemodynamics change with ageing (for example, resting blood flow falls¹²³ and perivascular space function diminishes^{47,124}). Studies of post-mortem samples from patients with stroke or AD, in which the age matching was more meaningful, showed that aberrant location of AQP4 on astrocyte end feet, astrocyte damage and BBB disruption were all associated with cognitive impairment^{121,125,126}, although these changes could be co-associations rather than causations.

[H1] Translation to the clinic

In clinical practice, we need to know what it means if many perivascular spaces are visible on MRI. As described above (See Insights from MRI above) [Au: OK?ok], high numbers of visible perivascular spaces on MRI, particularly at a young age, are an indicator of impaired brain health, although this observation is currently non-specific. [Au: Edited to avoid repetition. OK?ok] Interpretation of the finding and the action required depends on several factors, including the age of the patient, the indication for the scan, a history of hypertension, the patient's blood pressure, and the presence of other vascular risk factors, other features of SVD, such as WMH or microbleeds, cognitive impairment and systemic inflammation. These factors and disorders are all associated with perivascular spaces and therefore indicate a more detailed assessment of modifiable risk factors and appropriate interventions. For example, interventions might include blood pressure reduction if the patient has hypertension, or improvement of glycaemic control if the patient has diabetes mellitus.

However, much more research and knowledge is needed to make better use of information about perivascular spaces. For example, a better understanding of how many perivascular spaces are normal at different ages is required. Simpler methods to quantify perivascular spaces are needed to determine when perivascular spaces are exceeding normal number and size limits; these methods might be simpler visual rating scales or better computational methods. Much more data are required from long-term studies to determine the clinical relevance of perivascular spaces and the associated risks of future diseases — in particular stroke and dementia — when high numbers of visible perivascular spaces are identified, particularly in younger patients. Additionally, at least some improvement in our understanding of the implications of perivascular spaces could be obtained by further analysis of existing datasets that include young and middle-aged cohorts currently in long-term follow-up periods, or (where available) by 'big data' analysis of routine clinical MRI linked to central health databases. [Au: Edited sentence OK?ok] New MRI studies should also include a T2 volume sequence to optimize perivascular

space detection and should include long-term follow-up for cognitive and vascular disease outcomes.

[H1] Conclusions and perspectives

Our understanding of perivascular spaces has improved, but remains patchy. Fluid and metabolite management in the brain is difficult to research because a delicate milieu is necessary for neurons to function, making it difficult to perform experiments that do not inadvertently upset this delicate balance. We do know that increased visibility of perivascular spaces on MRI in humans is a marker of age, several vascular risk factors, inflammation and BBB leakage, that the increase in visibility occurs in conjunction with other features of SVD and is associated with cognitive decline and, by extrapolation from post-mortem and rodent studies, that increased visibility of perivascular spaces is a marker of impaired flushing of brain ISF.

However, several observations remain unexplained (Box 1). For example, the perivascular spaces seen on MRI in humans are most likely periarteriolar, CSF tracer [Au: Addition of “tracer” OK?ok] uptake is most pronounced in periarteriolar spaces in rodents and tracers injected intravenously or intrastrially are seen around arterioles in humans and rodents, yet ISF ~~should~~ is thought to [Au: Please clarify what you mean by “should”. For what reason “should” they drain via perivenular spaces?] drain via perivenular spaces. Few perivenular spaces have been described or imaged, possibly owing to experimental challenges. [Au: Removed details to avoid repetition from earlier in the article and to streamline.ok] Furthermore, ethical issues preclude injection of radioactive tracers, such as gadolinium, into CSF in humans for research purposes — the few studies that have been conducted with this contrast agent were in patients already undergoing investigation for abnormal CSF dynamics^{92,138}.

Where the perivenular fluid drains to is also unclear, but rodent studies suggest that it flows via the subpial (and/or subarachnoid) spaces to dural and nasal lymphatics and cervical lymph nodes^{8,139}. A lack of studies and major technical challenges mean that only limited evidence is available for the existence of dural lymphatics in humans, for their connections [Au: Connections of what to cervical lymph nodes? Please specify] to cervical lymph nodes, and for the relative contributions of each route to intracranial fluid drainage in physiological and pathological conditions. The route by which excess A β generated in the brain is deposited in periarteriolar spaces and arteriole walls in CAA and AD [Au: Addition of “in CAA and AD” OK?ok] is unclear. In addition, AQP4 channels on astrocyte

end-feet are thought to be important in promoting ISF flow through the brain, yet little is known of AQP4 function or abnormalities in the human brain in cognitive or vascular disease states^{125,140}.

The gaps in our knowledge about perivascular spaces and their relationship with fluid drainage systems have [Au: Addition made to be more specific about what has major implications. OK?ok] major implications for future research. Differences between rodent and human brains mean that the more studies are done in humans, the more relevant the results will be. However, animal studies enable use of invasive manipulations that are not possible in humans but are likely to shed light on mechanistic aspects of perivascular spaces and their relationship with disease.

[Au: Sentence shortened to streamline. OK?ok] New imaging approaches that enable reliable visualization of fluid flow directly to avoid the limitations of tracers would be helpful. Current in vivo MRI methods enable measurement of the shape, size, number and location of perivascular spaces, as well as intracranial vascular and CSF pulsatility, flow, CVR and BBB function. Large-scale longitudinal studies are now required to understand how perivascular space dysfunction starts, how this dysfunction affects ISF balance, how it leads to brain damage, and whether and how this brain damage [Au: this process or this brain damage?] can be reversed. Studies are needed to determine the relative roles of dural lymphatics versus other pathways in drainage of fluid from the brain, ~~and the role that cervical lymph nodes have in maintaining the brain's immune competence.~~ [Au: The concept of immune competence has not been mentioned previously, and it's therefore not clear how it relates to cervical lymph nodes here. I would suggest removing this or re-wording it to avoid introducing a new concept right at the end of the article.deleted] Finally, through large, longitudinal studies in diverse populations, we need clarification of the clinical implications of increased perivascular space visibility on brain MRI as a possible early warning sign of vascular dysfunction, neurodegenerative and neuroinflammatory disorders. [Au: How might this clarification be achieved? How will this clarification help clinical practice? Please explain]

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Related links

Fondation Leducq Transatlantic Network of Excellence on the Role of the Perivascular Space in Cerebral Small Vessel Disease: www.small-vessel-disease.org

Harmonising Brain Imaging Methods for Vascular Contributions to Neurodegeneration (HARNESS) <https://harness-neuroimaging.org/>

Key points [Au: The key points do not count as a box that can be cited in the article, so I have removed the numbering of this as a box.ok]

- Visible perivascular spaces on MRI increase in number with age, vascular risk factors (particularly hypertension) and other features of small vessel disease, indicating that they are clinically relevant.
- Perivascular space dilation on MRI is a marker of perivascular space dysfunction and, by implication from preclinical studies, [Au: Addition OK?ok] impairment of normal brain fluid and waste clearance and microvascular dysfunction.
- Perivascular spaces can be quantified using visual scores of perivascular spaces in standard brain regions, and now also with computational measures of perivascular space count, volume, length, width, sphericity and orientation.
- Experimental models show that perivascular spaces are important conduits for uptake of cerebrospinal fluid to flush interstitial fluid and clear metabolic waste; these processes might increase during sleep.
- The relative importance of different drainage routes from perivascular spaces in humans remains to be determined.

Box 1 | Big questions related to perivascular spaces

1. Does interstitial fluid (ISF) drain via perivenular spaces and, if so, where to?
2. Once out of the brain, what proportion of cerebrospinal fluid (CSF) and ISF reach sub-pial and sub-arachnoid spaces, and what proportion drains via dural lymphatics or arachnoid granulations and villi?
3. If ISF drains via perivenular spaces, why does amyloid- β and other debris collect in periaarteriolar spaces?
4. In humans, are most perivascular spaces that are visible on MRI periaarteriolar, perivenular, or both; if both, is it possible to distinguish these spaces?
5. In humans, to what extent do vascular pulsation and/or respiratory motion facilitate fluid movement through perivascular spaces?
6. In humans, does aquaporin 4 facilitate uptake of CSF into perivascular spaces and the clearance of ISF?
7. In humans, does perivascular space function differ during sleep and wakefulness?

Review criteria

We used recent systematic reviews where available and updated their contents. We searched the literature from the mid-1800s to the present for papers on 'perivascular spaces', 'glymphatics', 'Virchow–Robin spaces', 'small vessel disease', 'cerebrospinal fluid', 'cerebral blood flow', 'white matter hyperintensities', 'lacunes', 'microbleeds', 'siderosis', 'stroke', 'dementia', 'cognition', 'magnetic resonance imaging', '2-photon imaging', 'electron microscopy' and 'immunohistochemistry'. Where available, we used recent systematic reviews and updated their contents. We looked for additional relevant papers in reference lists of review articles and research papers. Our approach was not systematic owing to the breadth of the field, but we aimed to capture key papers in the field. We discussed and debated at length the historical and recent findings in our Leducq research network.

Figure 1 | Perivascular spaces visualized with MRI in humans. a | Typical perivascular spaces in the temporo-occipital region as they appear on T2-weighted MRI. The spaces appear as thin white lines that extend from the lateral ventricle towards the cortex. [Au: I think some arrows would be helpful to indicate the lines you are referring to for those who are not familiar with these images.] b | [Au: I would recommend the two bottom panels of part a are moved to a separate part of the figure for clarity of the legend. OK?ok] Schematic representations of perivascular spaces as they look on T2-weighted, T1-weighted and fluid attenuation inversion recovery (FLAIR) images [Au: I suggest labelling the two appearances in the figure rather than explaining them in the legend.ok] (adapted from STandards for Reporting Vascular changes on nEuroimaging, STRIVE⁷). [Au: Please note that this article is not published under a creative commons licence and the copyright therefore lies with Lancet. Unfortunately, Lancet do not allow modifications of figures (such as using part of a panel, as is the case here), so we will not get permission to use this image. The schematics are not a problem because we can redraw them to look different, but we will unfortunately need to use an alternative MRI scan for part a to illustrate the appearance of typical perivascular spaces.] c | T2-weighted MRI scans that illustrate different extents of perivascular space visibility in the basal ganglia (top row) and the centrum semiovale (bottom row). Visual rating scores (1 to 4) [Au: Please define EPVS – perivascular spaces] are indicated at the bottom⁴³. [Au: Please clarify the source of the images in this part of the figure. The bottom row seem to be from reference 31, but I cannot see where the images in the top row are from – they do not seem to be in reference 43 that is cited or in reference 31. Are they previously unpublished? Both rows are from a teaching file that I use – not] d | Axial view of moderate (left) and severe (right) perivascular spaces identified computationally³¹. [Au: We generally try to avoid reproducing data like that in part d – the relationship perivascular spaces and WMHs is stated clearly enough in the text, so the data are not needed to illustrate this. I therefore suggest removal of part d.]

Kommenterede [WJ16]: Hmm. It was published open access. The funder paid a lot of money for this. Are you sure?

Figure 2 | High-field and conventional MRI of perivascular spaces in humans in vivo and at post-mortem. a | A coronal high-field (7T) T2-weighted [Au: OK?ok] MRI scan of perforating arterioles and perivascular spaces in the basal ganglia. The rectangle

illustrates the area visualized in part b. b | T2-weighted MRI (top) and T1-weighted MRI (middle) reveal the perivascular spaces that correspond with the arterioles visible with magnetic resonance angiography (bottom). [Au: Edited to point out the correspondence of the perivascular spaces with the vessel. OK?ok] The inferior ends of the perivascular spaces widen to join the basal cerebrospinal fluid (CSF; arrows). Reproduced from ref⁴.

c | T2-weighted 1.5T MRI images of the frontal lobe in two different patients. Perivascular spaces are visible approaching the cortex and are more dilated towards the inner edge of the cortex, but they are not visible passing through the cortex. [Au: Please add arrows to indicate the features mentioned.ok]

Figure 3 | Perivascular spaces and small vessel disease. a | [Au: For the first part of this figure, I have tried to adjust the legend so it is clearer how these images relate to amyloid deposition. However, where this figure is cited in the text, the discussion is specifically about conditions characterized by vascular deposition of A β , whereas this figure is in AD, so I am not sure this figure is an appropriate illustration of the point being made in the text. Please consider.] A haematoxylin and eosin-stained superior frontal gyrus and white matter section from an 80-year-old patient with AD (left) shows that enlarged perivascular spaces are associated with amyloid β deposition; numerous arterioles have enlarged perivascular spaces throughout the white matter that do not propagate through the cortex. The pale blotches in the white matter are areas of myelin rarefaction. In a section from a cognitively normal 74-year old individual who died of a non-brain disorder (right), no perivascular space enlargement is seen. A similar section from an 80-year-old patient with AD (right) shows that enlarged perivascular spaces are associated with amyloid- β deposition; numerous arterioles have enlarged perivascular spaces throughout the white matter that do not propagate through the cortex. The pale blotches in the white matter are areas of myelin rarefaction. Magnification: $\sim 2.5\times$. Reproduced from ref⁵⁶. d | Axial 1.5T brain MRI scans from 75-year-old individual with minor ischaemic stroke. T2-weighted MRI (left) reveals numerous perivascular spaces (circled) in the white matter running perpendicular to the cortex. Fluid attenuation inversion recovery (FLAIR) imaging (right) reveals white matter hyperintensities starting to form around the perivascular spaces, including some large white matter hyperintensities (lower circle) that look similar to linear and blotchy white matter rarefaction (as in part a).

Kommenterede [WJ17]: Note I don't have the amended figures but relative to the original figure, the order of the legend for the first pair of images was incorrect therefore switched. Need to check once see figs

Figure 4 | Relationship of perivascular spaces to arterioles and venules in humans. a | High-field, 7T MRI shows that perivascular spaces (small A, magnified in C) are spatially aligned with arterioles and not venules. B (magnified in D) shows 2D time of flight first echo angiography revealing arterioles and venules as hyperintense structures (open and solid arrows respectively). Small E shows 2D time of flight 3rd echo angiogram in which the venules become black (solid arrows). Comparing C, D and E shows that perivascular spaces correspond in location with arterioles and not venules. Reproduced from ref⁴. [Au: Please add more explanation of what each of the images shows and what the arrows indicate so it is clear how this shows the alignment with arteries and not venules.] b | [Au: Is part b necessary, given that part a already shows the lack of alignment with venules? Yes Please consider hope clearer now] Adjacent slices from a T2-weighted 1.5T MRI images (top and middle), [Au: If this figure is kept, please clarify what the different is between the top and middle images are. Are both the top and middle images needed?yes] and coregistered T2*-weighted 1.5T MRI image (bottom) showing that perivascular spaces (white arrows) do not align with venules (black arrows and arrowhead) but run parallel to the venules. [Au: Change to “venules” OK? If incorrect, please clarify.ok]

Formateret: Hævet skrift

Figure 5 | Anatomy of perivascular spaces around ~~in~~-basal and cortical perforating arterioles. Basal arterioles (left) have two meningeal membranes, whereas cortical arterioles (right) and all venules have only one meningeal membrane so are thought to communicate with the subpial rather than the subarachnoid space⁸². [Au: Which is thought to communicate with the subpial space? The cortical arterioles and venules Neither appear to communicate with the subpial space in the diagram, but both appear to communicate with the subarachnoid space. Actually on the left the space around the vessel is contiguous with the SPS whereas on the right the double membrane encloses a space that connects to the SAS Please clarify.] Reproduced from ref⁸²

[Au: Looking at where Figure 4 is cited in the text, it's not entirely clear to me what the main point of this figure is and what it tells us about perivascular spaces. Given that it's also a rodent experiment and that there are already several figures and the article is quite long, I would suggest omitting this figure, because I don't think it's essential. Please consider.] it would be a shame to omit this figure since it shows the CSF tracer being taken up into perivascular spaces from the CSF supporting the notion that flow is up the

PVS not down. Its very cutting edge. It could be retitled 'Uptake of CSF tracer into perivascular spaces' rather than glymphatics...

Figure 6 | Uptake of cerebrospinal fluid into perivascular spaces in rodents and humans. a | Mouse (left) and rat (right) coronal brain slices taken just anterior (top) and posterior (bottom) to the bregma 30 minutes after intracisternal infusion of a cerebrospinal fluid (CSF) tracer. The distribution of uptake into the cortex is similar between species. Uptake is less extensive in rats, possibly owing to its larger size. Imaging was performed while the rodent was ventral side down. Reproduced from ref¹⁴¹. b | Series of coronal (top) and axial (bottom) T1-weighted MRI scans of one patient undergoing investigation for a neurological disorder by injection of gadolinium into the lumbar spinal CSF. The patient remained in the supine position throughout. Note the higher uptake of gadolinium in the cortex relative to that in the basal ganglia. Reproduced from ref⁹².

[Au: I don't think the aquaporin 4 discussion is sufficiently central to the article to justify inclusion of data on its role, so I would suggest part c is omitted.] shame. It would help to balance the pros and cons of PVS function

[Au: In relation to part d, please see my comments on the section that this image relates to. As for the aquaporin 4 image, I don't think the BBB leakage discussion justifies a figure to illustrate it, so I'd suggest that part d is omitted.] ok

Figure 7 | Movement of fluid in perivascular spaces.a | Trajectories of microspheres that had were infused into the cisterna magna and visualized through a cranial window (inset) using 2-photon imaging in live mice. The particles were transported primarily in large perivascular spaces. b | Fluorescent dextran in the CSF (green) indicates the location of perivascular spaces in the location shown in part a. The spaces are primarily around pial arteries.

[Au: I don't think the points illustrated in parts c and d warrant inclusion of figures, so I suggest only parts a and b are included to illustrate the flow. OK?]

Actually the point about change in PVS shape and collapse post mortem is important since it may explain why some expts suggest (probably erroneously) that tracer moves along basement membranes – part c illustrates how the tracer sticks to the vessel wall whereas in vivo it does not. Part d shows how the pvs collapses ex vivo. I think this is quite important to understanding sources of disagreement.

Reproduced from ref¹³

Glossary terms [Au: I have copied over terms from your glossary that I think will be helpful – I think the others were either terms that are very familiar to our audience or are adequately and appropriately explained within the text. ok]

Perivascular spaces

A space or potential space around arterioles, capillaries and venules in the brain, along which fluid or particles can pass. Not restricted to Virchow–Robin spaces.

Virchow–Robin spaces

Macroscopic spaces originally identified in post-mortem brain specimens surrounding the perforating vessels in the basal ganglia and hemispheric white matter. Thought to correspond to the perivascular spaces that are visible on brain MRI. [Au: Edited version OK? ok]

Lacune

A small hole in the deep grey or white matter, often the sequelae of a small deep lacunar infarct but commonly found in persons with no prior symptoms; increases with age, associated with cognitive decline, part of the spectrum of small vessel disease.

Optimal mass transport

A method of analysing the passage of a fluid (for example, contrast agent) through a volume, (for example, the intracranial cavity).